

Statistics & Probability

ANSWER KEY



Mean, median, and mode

- 1 Find the mean of 4, 8, 6, 10, 7.

$$\text{Sum} = 35. \text{ Mean} = \frac{35}{5} = 7.$$

- 2 Find the median of 3, 9, 5, 1, 7.

Sorted: 1, 3, 5, 7, 9. Median = 5.

- 3 Find the mode of 2, 4, 4, 6, 8, 4, 2.

4 appears three times, more than any other value, so the mode is 4.

- 4 A data set has values 8, 12, 10, 15, 10, 14, 11. Find the mean, median, and mode.

Sum = 80, mean = $\frac{80}{7} \approx 11.43$. Sorted: 8, 10, 10, 11, 12, 14, 15; median (4th value) = 11. Mode = 10 (appears twice).

Range and quartiles

- 5 Find the range of 5, 12, 8, 20, 3.

$$\text{Range} = 20 - 3 = 17.$$

- 6 For the data set 2, 4, 5, 7, 8, 9, 12, find Q_1 and Q_3 .

Median (4th value) = 7. Lower half 2, 4, 5: $Q_1 = 4$. Upper half 8, 9, 12: $Q_3 = 9$.

- 7 Find the interquartile range (IQR) for the same data set.

$$\text{IQR} = Q_3 - Q_1 = 9 - 4 = 5.$$

Variance and standard deviation

- 8 Find the variance of 2, 4, 6, 8, 10, using $\sigma^2 = \frac{\sum (x - \mu)^2}{n}$.

Mean = 6. Squared deviations: 16, 4, 0, 4, 16, sum = 40. Variance = $\frac{40}{5} = 8$.

- 9 Find the standard deviation of the same data set, rounded to two decimal places.

$$\sigma = \sqrt{8} \approx 2.83.$$

- 10 A data set has variance 25. State its standard deviation.

$$\sigma = \sqrt{25} = 5.$$

Frequency tables

- 11 A frequency table shows value 1 with frequency 3, value 2 with frequency 5, and value 3 with frequency 2. Find the mean.

$$\text{Sum} = 1(3) + 2(5) + 3(2) = 3 + 10 + 6 = 19. \text{ Total frequency} = 10. \text{ Mean} = \frac{19}{10} = 1.9.$$

- 12 Using the same frequency table, find the mode.

Value 2 has the highest frequency (5), so the mode is 2.

- 13 Using the same frequency table, find the median.

There are 10 values in order 1, 1, 1, 2, 2, 2, 2, 2, 3, 3. Median = $\frac{2+2}{2} = 2$ (average of the 5th and 6th values).

Basic probability

- 14 A fair six-sided die is rolled. Find $P(\text{number} > 4)$.

$$\text{Favorable outcomes } \{5, 6\}: P = \frac{2}{6} = \frac{1}{3}.$$

- 15 A bag contains 5 red and 3 blue marbles. Find $P(\text{blue})$.

$$P(\text{blue}) = \frac{3}{8}.$$

- 16 A card is drawn from a standard 52-card deck. Find $P(\text{king})$.

$$P(\text{king}) = \frac{4}{52} = \frac{1}{13}.$$

- 17 Two fair coins are flipped. Find $P(\text{both heads})$.

$$P = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}.$$

Complementary and mutually exclusive events

- 18 If $P(A) = 0.35$, find $P(\text{not } A)$.

$$P(\text{not } A) = 1 - 0.35 = 0.65.$$

- 19 A die is rolled. Events $A = \{\text{even}\}$ and $B = \{\text{rolling a 5}\}$. Are A and B mutually exclusive?

Yes — $\{2, 4, 6\}$ and $\{5\}$ share no outcomes.

- 20 If $P(A) = 0.4$ and $P(B) = 0.25$, and A, B are mutually exclusive, find $P(A \text{ or } B)$.

$$P(A \text{ or } B) = 0.4 + 0.25 = 0.65.$$

Independent events

- 21 Two independent events have $P(A) = 0.6$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

$$P(A \text{ and } B) = 0.6 \times 0.5 = 0.3.$$

- 22 A fair coin is flipped twice. Find $P(\text{first heads and second tails})$.

$$P = 0.5 \times 0.5 = 0.25.$$

- 23 A card is drawn from a deck, replaced, then a second card is drawn. Find $P(\text{both are aces})$.

$$P = \frac{4}{52} \times \frac{4}{52} = \frac{1}{13} \times \frac{1}{13} = \frac{1}{169}.$$

Conditional probability

- 24 In a class, $P(\text{math}) = 0.7$ and $P(\text{physics and math}) = 0.4$. Find $P(\text{physics} \mid \text{math})$.

$$P(\text{physics} \mid \text{math}) = \frac{0.4}{0.7} \approx 0.571.$$

- 25 A box has 10 balls: 6 red and 4 blue. Two balls are drawn without replacement. Find $P(\text{2nd is blue} \mid \text{1st is blue})$.

$$\text{After removing one blue ball, 3 of the remaining 9 balls are blue: } P = \frac{3}{9} = \frac{1}{3}.$$

- 26 Given $P(A) = 0.5$, $P(B) = 0.3$, $P(A \text{ and } B) = 0.15$, find $P(A \mid B)$.

$$P(A \mid B) = \frac{0.15}{0.3} = 0.5.$$

Counting: permutations and combinations

- 27 In how many ways can 4 books be arranged on a shelf?

$$4! = 24 \text{ ways.}$$

- 28 How many ways can a committee of 3 be chosen from 8 people (order doesn't matter)?

$$\binom{8}{3} = \frac{8!}{3!5!} = 56.$$

- 29 A code uses 3 distinct digits from 0–9, where order matters. How many possible codes are there?

$$10 \times 9 \times 8 = 720.$$

The five-number summary

- 30 For the data set 6, 10, 14, 18, 22, 26, 30, find the five-number summary (minimum, Q_1 , median, Q_3 , maximum).

Median (4th value) = 18. Lower half 6, 10, 14: $Q_1 = 10$. Upper half 22, 26, 30: $Q_3 = 26$. Summary: 6, 10, 18, 26, 30.

- 31 Find the IQR for the same data set.

$$\text{IQR} = Q_3 - Q_1 = 26 - 10 = 16.$$

- 32 Outlier boundaries are $Q_1 - 1.5 \times \text{IQR}$ and $Q_3 + 1.5 \times \text{IQR}$. Find both boundaries for the same data set.

$$\text{Lower: } 10 - 1.5(16) = 10 - 24 = -14. \text{ Upper: } 26 + 1.5(16) = 26 + 24 = 50.$$

Reading data from a bar chart

- 33 The bar chart shows the number of books sold each day at a bookstore over five days.

a Find the total number of books sold over the five days. Total = $12 + 18 + 9 + 15 + 21 = 75$ books.

b Find the mean number of books sold per day. Mean = $\frac{75}{5} = 15$ books per day.

Word problems: statistics and probability in context

- 34 This question has two parts. A survey of 200 students found that 120 like math and 90 like science, while 50 like both subjects.

a Find the probability that a randomly selected student likes math or science, using
 $P(M \cup S) = P(M) + P(S) - P(M \cap S)$. $P(M) = \frac{120}{200} = 0.6$, $P(S) = \frac{90}{200} = 0.45$, $P(M \cap S) = \frac{50}{200} = 0.25$.
 $P(M \cup S) = 0.6 + 0.45 - 0.25 = 0.8$.

b Find the probability that a student likes neither subject. $P(\text{neither}) = 1 - 0.8 = 0.2$.

- 35** This question has two parts. A quality-control inspector finds that 2% of items from a production line are defective. A random sample of 3 items is tested, and defects occur independently.
- a** Find the probability that all three items are defective. $P(\text{all defective}) = 0.02^3 = 0.000008$.
 - b** Find the probability that none of the three items are defective. $P(\text{none defective}) = 0.98^3 \approx 0.9412$.
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MCQ — Mean, median, and mode

- 36** Find the mean of 5, 9, 7, 11, 8.

- a.** 8 ✓
- b.** 7
- c.** 40
- d.** 9

Sum = 40; mean = $40 \div 5 = 8$.

- 37** Find the median of 4, 10, 6, 2, 8.

- a.** 5
- b.** 8
- c.** 6 ✓
- d.** 4

Sorted: 2, 4, 6, 8, 10; median = 6.

- 38** Find the mode of 3, 5, 5, 7, 9, 5, 3.

- a.** 5 ✓
- b.** 4
- c.** 3
- d.** 7

5 appears 3 times, more than any other value.

39 A data set has values 9, 13, 11, 16, 11, 15, 12. Find the mean, median, and mode.

- a. Mean = 13, median = 12, mode = 11
- b. Mean = 12.43, median = 11, mode = 12
- c. Mean $\approx$ 12.43, median = 12, mode = 11 ✓
- d. Mean $\approx$ 12.43, median = 13, mode = 11

Sum = 87, mean = $87 \div 7 \approx 12.43$. Sorted: 9, 11, 11, 12, 13, 15, 16; median = 12; mode = 11.

40 Find the mean of 12, 18, 15, 21, 14.

- a. 15
- b. 16 ✓
- c. 80
- d. 18

Sum = 80; mean = $80 \div 5 = 16$.

MCQ — Range and quartiles

41 Find the range of 6, 14, 9, 22, 4.

- a. 22
- b. 16
- c. 18 ✓
- d. 20

$22 - 4 = 18$.

42 For the data set 3, 5, 6, 8, 9, 10, 13, find Q_1 and Q_3 .

a. $Q_1 = 5, Q_3 = 9$

b. $Q_1 = 5, Q_3 = 10$ ✓

c. $Q_1 = 3, Q_3 = 13$

d. $Q_1 = 6, Q_3 = 9$

Lower half 3, 5, 6: $Q_1 = 5$; upper half 9, 10, 13: $Q_3 = 10$.

43 Find the interquartile range (IQR) for the same data set.

a. 10

b. 8

c. 3

d. 5 ✓

$$IQR = Q_3 - Q_1 = 10 - 5 = 5.$$

44 Find the range of 18, 25, 12, 30, 9.

a. 18

b. 16

c. 21 ✓

d. 30

$$30 - 9 = 21.$$

MCQ — Variance and standard deviation

45 Find the variance of 3, 5, 7, 9, 11, using $\sigma^2 = \frac{\sum(x - \mu)^2}{n}$.

- a. 4
- b. 2.83
- c. 8 ✓
- d. 40

Mean = 7; deviations squared sum to 40; variance = $40 \div 5 = 8$.

46 Find the standard deviation of the same data set, rounded to two decimal places.

- a. 1.41
- b. 4.00
- c. 2.83 ✓
- d. 8.00

$$\sqrt{8} \approx 2.83.$$

47 A data set has variance 36. State its standard deviation.

- a. 6 ✓
- b. 12
- c. 18
- d. 1,296

$$\sqrt{36} = 6.$$

48 Find the variance of 4, 8, 12, 16, 20, using $\sigma^2 = \frac{\sum(x - \mu)^2}{n}$.

- a. 160
- b. 5.66
- c. 16
- d. 32 ✓

Mean = 12; deviations squared sum to 160; variance = $160 \div 5 = 32$.

MCQ — Frequency tables

49 A frequency table shows value 1 with frequency 4, value 2 with frequency 6, and value 3 with frequency 5. Find the total number of data points.

- a. 10
- b. 15 ✓
- c. 3
- d. 6

$$4 + 6 + 5 = 15.$$

50 Using the same frequency table, find the mode.

- a. 6
- b. 3
- c. 2 ✓
- d. 1

Value 2 has the highest frequency (6).

51 Using the same frequency table, find the median.

- a. 1.5
- b. 1
- c. 2 ✓
- d. 3

With 15 values, the median is the 8th, which falls in the value-2 group.

52 Using the same frequency table, find the mean.

a. 2.33

b. ≈ 2.07 ✓

c. 2.00

d. 1.87

$$\text{Sum} = 1(4) + 2(6) + 3(5) = 31; \text{mean} = 31 \div 15 \approx 2.07.$$

53 The bar chart shows a frequency table. Find the total number of data points.

a. 15 ✓

b. 6

c. 3

d. 10

$$4 + 6 + 5 = 15.$$

MCQ — Basic probability

54 A fair six-sided die is rolled. Find $P(\text{number} < 3)$.

a. $\frac{1}{6}$

b. $\frac{1}{2}$

c. $\frac{1}{3}$ ✓

d. $\frac{2}{3}$

$$\{1, 2\}: \frac{2}{6} = \frac{1}{3}.$$

- 55 A bag contains 7 red and 4 blue marbles. Find $P(\text{red})$.

- a. $\frac{7}{11}$ ✓
- b. $\frac{1}{11}$
- c. $\frac{4}{11}$
- d. $\frac{7}{4}$

7 out of 11 total.

- 56 A card is drawn from a standard 52-card deck. Find $P(\text{queen})$.

- a. $\frac{4}{13}$
- b. $\frac{1}{4}$
- c. $\frac{1}{13}$ ✓
- d. $\frac{1}{52}$

4 queens out of 52: $\frac{1}{13}$.

- 57 Two fair coins are flipped. Find $P(\text{at least one tail})$.

- a. $\frac{1}{2}$
- b. $\frac{3}{4}$ ✓
- c. $\frac{1}{4}$
- d. $\frac{1}{3}$

$P(\text{no tails}) = \frac{1}{4}$, so $P(\text{at least one}) = \frac{3}{4}$.

- 58 A fair six-sided die is rolled. Find $P(\text{multiple of 3})$.

- a. $\frac{2}{5}$
- b. $\frac{1}{3}$ ✓
- c. $\frac{1}{2}$
- d. $\frac{1}{6}$

$\{3, 6\}$: $\frac{2}{6} = \frac{1}{3}$.

MCQ — Complementary and mutually exclusive events

59 If $P(A) = 0.28$, find $P(\text{not } A)$.

- a. 0.28
- b. 1.28
- c. 0.5
- d. 0.72 ✓

$$1 - 0.28 = 0.72.$$

60 A die is rolled. Events $A = \{\text{odd}\}$ and $B = \{\text{rolling a } 4\}$. Are A and B mutually exclusive?

- a. Yes ✓
- b. Only sometimes
- c. No
- d. Cannot be determined

$A = \{1, 3, 5\}$ and $B = \{4\}$ share no outcomes: yes.

61 If $P(A) = 0.3$ and $P(B) = 0.45$, and A, B are mutually exclusive, find $P(A \text{ or } B)$.

- a. 0.75 ✓
- b. 0.15
- c. 1.05
- d. 0.135

$$0.3 + 0.45 = 0.75.$$

62 If $P(A) = 0.6$, find $P(\text{not } A)$.

- a. 1.6
- b. 0.4 ✓
- c. 0.5
- d. 0.6

$$1 - 0.6 = 0.4.$$

MCQ — Independent events

63 Two independent events have $P(A) = 0.7$ and $P(B) = 0.4$. Find $P(A \text{ and } B)$.

- a. 1.1
- b. 0.28 ✓
- c. 0.3
- d. 0.175

$$0.7 \times 0.4 = 0.28.$$

64 A fair coin is flipped twice. Find $P(\text{both heads})$.

- a. $\frac{1}{3}$
- b. 1
- c. $\frac{1}{4}$ ✓
- d. $\frac{1}{2}$

$$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}.$$

65 A card is drawn from a deck, replaced, then a second card is drawn. Find $P(\text{both are kings})$.

a. $\frac{2}{13}$

b. $\frac{1}{26}$

c. $\frac{1}{13}$

d. $\frac{1}{169}$ ✓

$$\left(\frac{4}{52}\right)^2 = \frac{1}{169}.$$

66 Two independent events have $P(A) = 0.5$ and $P(B) = 0.8$. Find $P(A \text{ and } B)$.

a. 0.65

b. 1.3

c. 0.3

d. 0.4 ✓

$$0.5 \times 0.8 = 0.4.$$

MCQ — Conditional probability

67 In a class, $P(\text{science}) = 0.65$ and $P(\text{math and science}) = 0.3$. Find $P(\text{math} \mid \text{science})$.

a. 0.65

b. 0.195

c. ≈ 0.46 ✓

d. 0.3

$$\frac{0.3}{0.65} \approx 0.46.$$

- 68 A box has 12 balls: 7 red and 5 blue. Two balls are drawn without replacement. Given the first is blue, find $P(\text{second is blue})$.

- a. $\frac{5}{11}$
- b. $\frac{4}{11}$ ✓
- c. $\frac{5}{12}$
- d. $\frac{4}{12}$

After removing one blue, 4 blue remain out of 11 total.

- 69 Given $P(A) = 0.6$, $P(B) = 0.4$, $P(A \text{ and } B) = 0.2$, find $P(A | B)$.

- a. 0.3
- b. 0.33
- c. 0.8
- d. 0.5 ✓

$$\frac{0.2}{0.4} = 0.5.$$

- 70 Given $P(A) = 0.45$, $P(B) = 0.5$, $P(A \text{ and } B) = 0.2$, find $P(B | A)$.

- a. 0.4
- b. 0.09
- c. 0.5
- d. ≈ 0.44 ✓

$$\frac{0.2}{0.45} \approx 0.44.$$

MCQ — Counting: permutations and combinations

71 In how many ways can 5 books be arranged on a shelf?

- a. 25
- b. 120 ✓
- c. 20
- d. 60

$$5! = 120.$$

72 How many ways can a committee of 4 be chosen from 9 people (order doesn't matter)?

- a. 126 ✓
- b. 3,024
- c. 362,880
- d. 36

$$\binom{9}{4} = 126.$$

73 A code uses 4 distinct digits from 0–9, where order matters. How many possible codes are there?

- a. 10,000
- b. 40
- c. 210
- d. 5,040 ✓

$$10 \times 9 \times 8 \times 7 = 5,040.$$

74 In how many ways can 6 people be arranged in a line?

- a. 720 ✓
- b. 120
- c. 36
- d. 6

$$6! = 720.$$

MCQ — The five-number summary

75 For the data set 5, 9, 13, 17, 21, 25, 29, find the five-number summary (minimum, Q_1 , median, Q_3 , maximum).

- a. 9, 13, 17, 21, 25
- b. 5, 9, 17, 25, 29 ✓
- c. 5, 13, 17, 21, 29
- d. 5, 9, 13, 25, 29

Min = 5; $Q_1 = 9$; median = 17; $Q_3 = 25$; max = 29.

76 Find the IQR for the same data set.

- a. 8
- b. 24
- c. 16 ✓
- d. 20

$IQR = Q_3 - Q_1 = 25 - 9 = 16$.

77 Outlier boundaries are $Q_1 - 1.5 \times IQR$ and $Q_3 + 1.5 \times IQR$. Find both boundaries for the same data set.

- a. -15 and 49 ✓
- b. 9 and 25
- c. -15 and 40
- d. -24 and 40

$9 - 1.5(16) = -15$; $25 + 1.5(16) = 49$.

78 For the data set 10, 15, 20, 25, 30, 35, 40, find the median.

- a. 25 ✓
- b. 30
- c. 20
- d. 27.5

With 7 values sorted, the median is the 4th: 25.

MCQ — Reading data from a bar chart

79 The bar chart shows the number of books sold each day at a bookstore over five days. Find the total books sold.

- a. 87 ✓
- b. 22
- c. 77
- d. 97

$$12 + 18 + 15 + 22 + 20 = 87.$$

80 Using the same bar chart, find the mean number of books sold per day.

- a. 15
- b. 18
- c. 20
- d. 17.4 ✓

$$87 \div 5 = 17.4.$$

81 Using the same bar chart, find the range of books sold.

- a. 12
- b. 22
- c. 8
- d. 10 ✓

$$22 - 12 = 10.$$

82 Using the same bar chart, on which day were the most books sold?

- a. Wednesday
- b. Thursday ✓
- c. Tuesday
- d. Friday

Thursday shows the tallest bar, 22 books.

MCQ — Word problems: statistics and probability in context

83 A survey of 250 students found that 150 like math and 100 like science, with 40 liking both. Find the probability that a randomly selected student likes math or science.

- a. 0.84 ✓
- b. 0.6
- c. 0.16
- d. 1.0

$$\frac{150 + 100 - 40}{250} = \frac{210}{250} = 0.84.$$

84 A quality-control inspector finds that 3% of items from a production line are defective. If 500 items are inspected, how many are expected to be defective?

a. 150

b. 15 ✓

c. 3

d. 30

$$0.03 \times 500 = 15.$$

85 In a survey of 180 people, 108 prefer coffee over tea. Find the probability, as a percent, that a randomly selected person prefers coffee.

a. 60% ✓

b. 50%

c. 108%

d. 40%

$$\frac{108}{180} = 60\%.$$

Multiple choice — easy

86 Find the mean of 11, 1, 2, 8, 18.

a. 7

b. 40

c. 8 ✓

d. 9

$$\text{Sum} = 40. \text{ Mean} = \frac{40}{5} = 8.$$

87 Find the mean of 2, 6, 1, 13, 8.

a. 30

b. 6 ✓

c. 7

d. 5

$$\text{Sum} = 30. \text{Mean} = \frac{30}{5} = 6.$$

88 Find the mean of 2, 22, 7, 12, 27.

a. 14 ✓

b. 13

c. 15

d. 70

$$\text{Sum} = 70. \text{Mean} = \frac{70}{5} = 14.$$

89 Find the mean of 17, 30, 23, 3, 7.

a. 16 ✓

b. 17

c. 80

d. 15

$$\text{Sum} = 80. \text{Mean} = \frac{80}{5} = 16.$$

90 Find the mean of 19, 25, 16, 14, 16.

a. 90

b. 17

c. 18 ✓

d. 19

$$\text{Sum} = 90. \text{Mean} = \frac{90}{5} = 18.$$

91 A bag contains 4 red and 2 blue marbles. Find $P(\text{blue})$.

a. $\frac{1}{4}$

b. $\frac{1}{3}$ ✓

c. $\frac{1}{2}$

d. $\frac{2}{3}$

$$P(\text{blue}) = \frac{2}{6} = \frac{1}{3}.$$

92 A bag contains 6 red and 3 blue marbles. Find $P(\text{blue})$.

a. $\frac{3}{11}$

b. $\frac{2}{3}$

c. $\frac{4}{9}$

d. $\frac{1}{3}$ ✓

$$P(\text{blue}) = \frac{3}{9} = \frac{1}{3}.$$

93 Two fair coins are flipped. Find $P(\text{one head one tail})$.

a. $\frac{1}{4}$

b. $\frac{1}{2}$ ✓

c. $\frac{3}{4}$

d. $\frac{1}{8}$

$$P = \frac{2}{4} = \frac{1}{2}.$$

94 A fair six-sided die is rolled. Find $P(\text{number} < 3)$.

a. $\frac{2}{7}$

b. $\frac{1}{3}$ ✓

c. $\frac{2}{3}$

d. $\frac{1}{2}$

$$P = \frac{2}{6} = \frac{1}{3}.$$

95 A card is drawn from a standard 52-card deck. Find $P(\text{ace})$.

a. $\frac{3}{26}$

b. $\frac{12}{13}$

c. $\frac{1}{13}$ ✓

d. $\frac{4}{51}$

$$P(\text{ace}) = \frac{4}{52} = \frac{1}{13}.$$

96 Find the range of 1, 12, 25, 17, 19.

a. 24 ✓

b. 22

c. 26

d. 25

$$\text{Range} = 25 - 1 = 24.$$

97 Find the range of 28, 2, 14, 37, 32.

a. 35 ✓

b. 37

c. 38

d. 33

$$\text{Range} = 37 - 2 = 35.$$

98 Find the range of 14, 39, 4, 32, 20.

- a. 39
- b. 35 ✓
- c. 37
- d. 33

$$\text{Range} = 39 - 4 = 35.$$

99 Find the range of 33, 17, 7, 15, 18.

- a. 26 ✓
- b. 24
- c. 28
- d. 33

$$\text{Range} = 33 - 7 = 26.$$

100 Find the range of 2, 1, 19, 5, 8.

- a. 20
- b. 16
- c. 19
- d. 18 ✓

$$\text{Range} = 19 - 1 = 18.$$

Multiple choice — medium

101 Find the median of 2, 19, 28, 15, 50.

- a. 19 ✓
- b. 22
- c. 17
- d. 21

Sorted: 2, 15, 19, 28, 50. Median = 19.

102 Find the median of 16, 40, 3, 43, 15, 26, 50.

- a. 28
- b. 27
- c. 26 ✓
- d. 24

Sorted: 3, 15, 16, 26, 40, 43, 50. Median = 26.

103 Find the median of 22, 47, 37, 14, 7.

- a. 22 ✓
- b. 20
- c. 24
- d. 25

Sorted: 7, 14, 22, 37, 47. Median = 22.

104 Find the median of 22, 11, 24, 37, 22.

- a. 23
- b. 24
- c. 20
- d. 22 ✓

Sorted: 11, 22, 22, 24, 37. Median = 22.

105 Find the median of 22, 15, 1, 21, 39.

a. 21 ✓

b. 23

c. 19

d. 24

Sorted: 1, 15, 21, 22, 39. Median = 21.

106 Two independent events have $P(A) = 0.7$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

a. 1.2

b. 0.35 ✓

c. 0.3

d. 0.4

$P(A \text{ and } B) = 0.7 \times 0.5 = 0.35$.

107 Two independent events have $P(A) = 0.8$ and $P(B) = 0.4$. Find $P(A \text{ and } B)$.

a. 0.37

b. 0.32 ✓

c. 1.2

d. 0.27

$P(A \text{ and } B) = 0.8 \times 0.4 = 0.32$.

108 Two independent events have $P(A) = 0.8$ and $P(B) = 0.6$. Find $P(A \text{ and } B)$.

- a. 1.4
- b. 0.48 ✓
- c. 0.53
- d. 0.43

$$P(A \text{ and } B) = 0.8 \times 0.6 = 0.48.$$

109 Two independent events have $P(A) = 0.5$ and $P(B) = 0.8$. Find $P(A \text{ and } B)$.

- a. 0.45
- b. 1.3
- c. 0.4 ✓
- d. 0.35

$$P(A \text{ and } B) = 0.5 \times 0.8 = 0.4.$$

110 Two independent events have $P(A) = 0.7$ and $P(B) = 0.8$. Find $P(A \text{ and } B)$.

- a. 0.56 ✓
- b. 1.5
- c. 0.61
- d. 0.51

$$P(A \text{ and } B) = 0.7 \times 0.8 = 0.56.$$

111 If $P(A) = 0.8$, find $P(\text{not } A)$.

- a. 0.8
- b. 0.3
- c. 0.2 ✓
- d. 0.1

$$P(\text{not } A) = 1 - 0.8 = 0.2.$$

112 If $P(A) = 0.2$, find $P(\text{not } A)$.

- a. 0.7
- b. 0.9
- c. 0.8 ✓
- d. 0.2

$$P(\text{not } A) = 1 - 0.2 = 0.8.$$

113 If $P(A) = 0.4$, find $P(\text{not } A)$.

- a. 0.7
- b. 0.6 ✓
- c. 0.5
- d. 0.4

$$P(\text{not } A) = 1 - 0.4 = 0.6.$$

114 If $P(A) = 0.3$, find $P(\text{not } A)$.

- a. 0.7 ✓
- b. 0.3
- c. 0.8
- d. 0.6

$$P(\text{not } A) = 1 - 0.3 = 0.7.$$

115 If $P(A) = 0.4$ and $P(B) = 0.4$, and A, B are mutually exclusive, find $P(A \text{ or } B)$.

a. 0.8 ✓

b. 0.16

c. 0.0

d. 0.9

$$P(A \text{ or } B) = 0.4 + 0.4 = 0.8.$$

Multiple choice — hard

116 A data set has variance 100. State its standard deviation.

a. 9

b. 11

c. 100

d. 10 ✓

$$\sigma = \sqrt{100} = 10.$$

117 A data set has variance 36. State its standard deviation.

a. 36

b. 5

c. 6 ✓

d. 7

$$\sigma = \sqrt{36} = 6.$$

118 Find the variance of 1, 3, 5, 7, 9, using $\sigma^2 = \frac{\sum(x - \mu)^2}{n}$.

a. 16

b. 13

c. 8 ✓

d. 5

Mean = 5. Squared deviations: 16, 4, 0, 4, 16, sum = 40. Variance = $\frac{40}{5} = 8$.

119 A data set has variance 49. State its standard deviation.

a. 8

b. 7 ✓

c. 6

d. 49

$\sigma = \sqrt{49} = 7$.

120 Find the variance of 10, 20, 30, 40, 50, using $\sigma^2 = \frac{\sum(x - \mu)^2}{n}$.

a. 200 ✓

b. 205

c. 400

d. 30

Mean = 30. Squared deviations: 400, 100, 0, 100, 400, sum = 1000. Variance = $\frac{1000}{5} = 200$.

121 A box has 12 balls: 4 red and 8 blue. Two balls are drawn without replacement. Find $P(\text{2nd is red} \mid \text{1st is red})$.

a. $\frac{3}{11}$ ✓

b. $\frac{1}{3}$

c. $\frac{4}{11}$

d. $\frac{8}{11}$

After removing one red ball, 3 of the remaining 11 balls are red: $P = \frac{3}{11}$.

122 A box has 10 balls: 4 red and 6 blue. Two balls are drawn without replacement. Find $P(\text{2nd is red} \mid \text{1st is red})$.

a. $\frac{4}{9}$

b. $\frac{2}{3}$

c. $\frac{1}{3}$ ✓

d. $\frac{2}{5}$

After removing one red ball, 3 of the remaining 9 balls are red: $P = \frac{3}{9} = \frac{1}{3}$.

123 A box has 12 balls: 3 red and 9 blue. Two balls are drawn without replacement. Find $P(\text{2nd is red} \mid \text{1st is red})$.

a. $\frac{1}{4}$

b. $\frac{3}{11}$

c. $\frac{2}{11}$ ✓

d. $\frac{9}{11}$

After removing one red ball, 2 of the remaining 11 balls are red: $P = \frac{2}{11}$.

124 Given $P(A) = 0.4$ and $P(A \text{ and } B) = 0.2$, find $P(B | A)$.

- a. 0.4
- b. 0.6
- c. 0.5 ✓
- d. 2.0

$$P(B | A) = \frac{0.2}{0.4} = 0.5.$$

125 A box has 8 balls: 5 red and 3 blue. Two balls are drawn without replacement. Find $P(\text{2nd is red} | \text{1st is red})$.

- a. $\frac{4}{7}$ ✓
- b. $\frac{5}{7}$
- c. $\frac{3}{7}$
- d. $\frac{5}{8}$

After removing one red ball, 4 of the remaining 7 balls are red: $P = \frac{4}{7}$.

126 A code uses 3 distinct digits from 0 to 9, where order matters. How many possible codes are there?

- a. 710
- b. 730
- c. 1000
- d. 720 ✓

$$10 \times 9 \times 8 = 720.$$

127 How many ways can a committee of 2 be chosen from 6 people (order doesn't matter)?

- a. 17
- b. 30
- c. 13
- d. 15 ✓

$$\binom{6}{2} = \frac{6!}{2!(4)!} = 15.$$

128 How many ways can a committee of 3 be chosen from 7 people (order doesn't matter)?

- a. 33
- b. 210
- c. 37
- d. 35 ✓

$$\binom{7}{3} = \frac{7!}{3!(4)!} = 35.$$

129 In how many ways can 6 books be arranged on a shelf?

- a. 726
- b. 720 ✓
- c. 714
- d. 36

$$6! = 720 \text{ ways.}$$

130 How many ways can a committee of 3 be chosen from 9 people (order doesn't matter)?

- a. 82
- b. 504
- c. 86
- d. 84 ✓

$$\binom{9}{3} = \frac{9!}{3!(6)!} = 84.$$